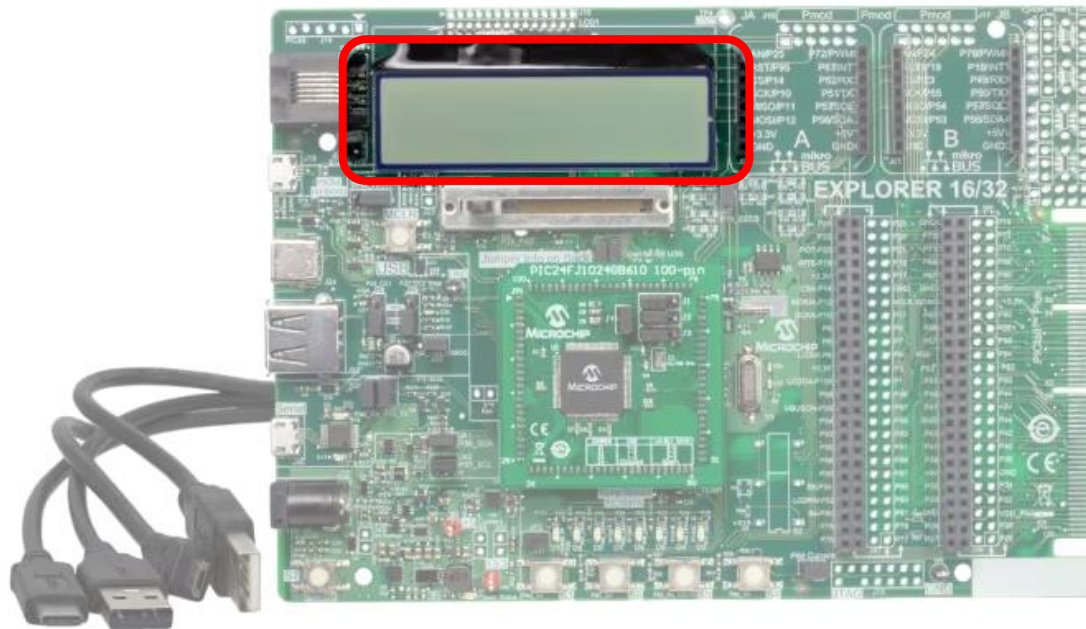
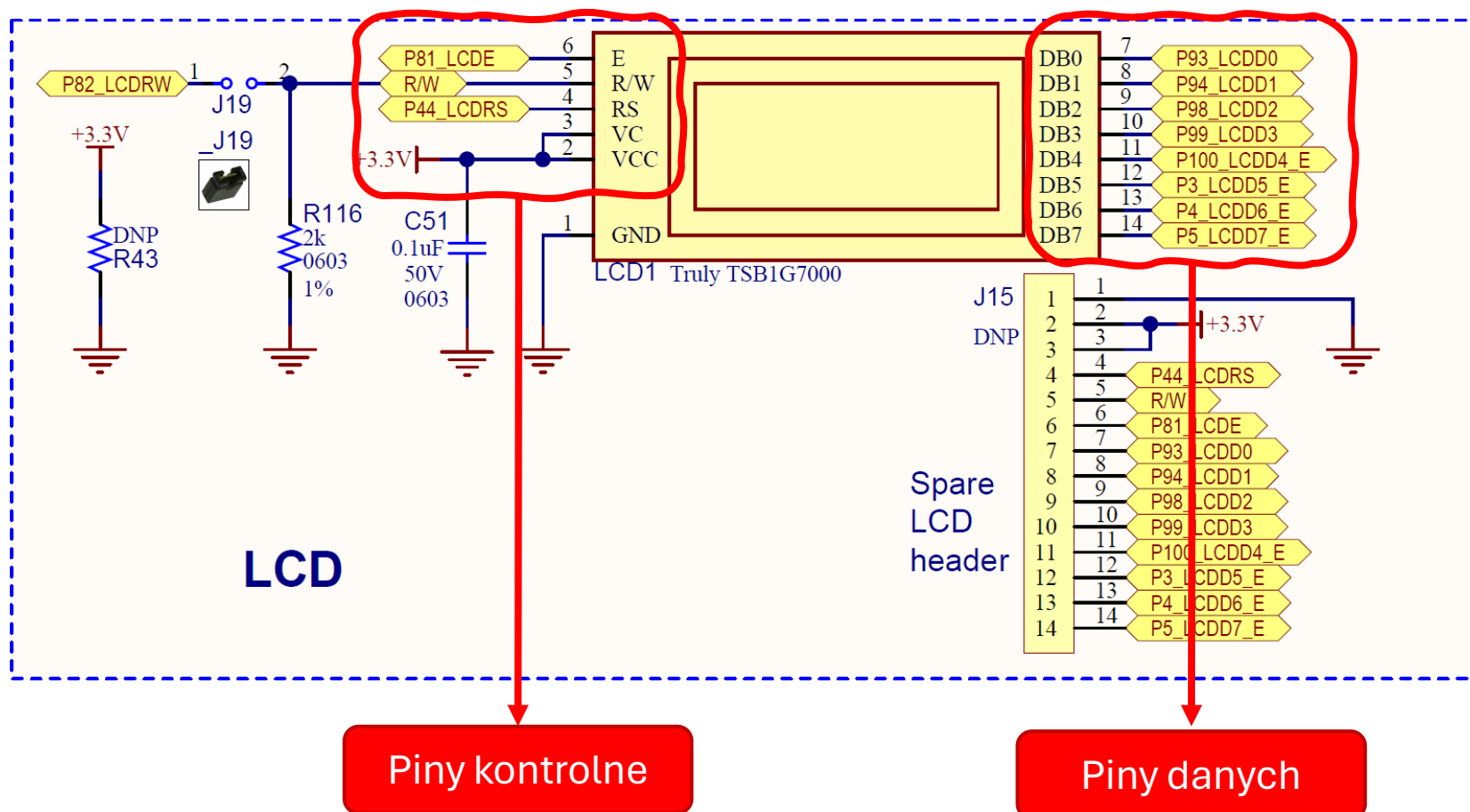


LCD

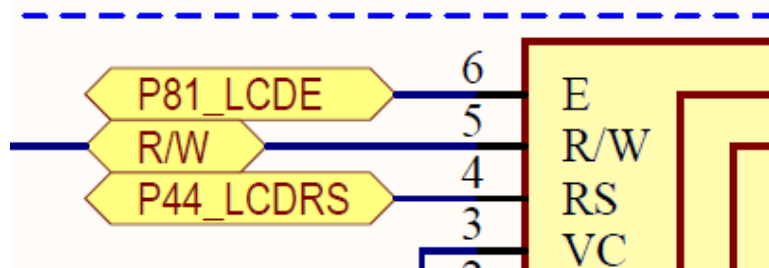


Dwie ścieżki zarządzania

piny kontrolne – piny danych

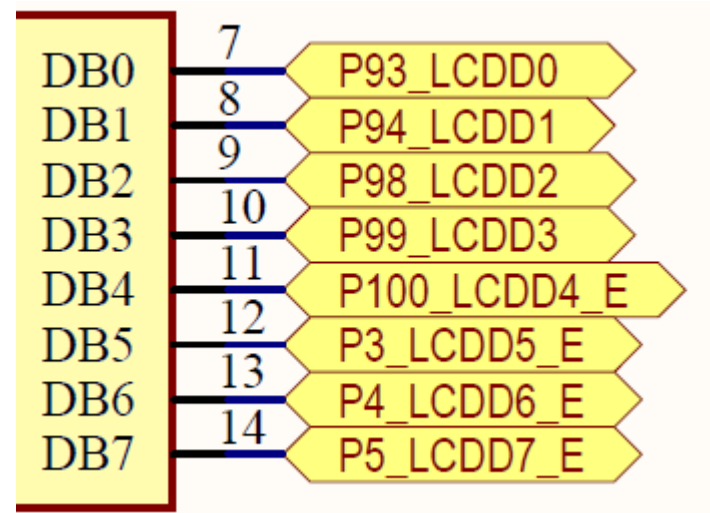


Adresacja pinów



E – P81 – port D pin 4
R/W – P82 – port D pin 5
RS – P44 – port B pin 15

E (enable) – aktywacja komunikacji
R/W (read/write) – odczyt/zapis
RS (register select) – wybór
pomiedzy przesyłaniem komend i
przesyłaniem danych



DB0-7 – P93-P5 – port E piny 0-7

Zarówno komendy jak i dane
wysyłane są szyną danych!
Steruje tym rejestr RS.

Komendy

zgodnie ze specyfikacją sterownika HD44780

Instruction	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Description
Clear display	0	0	0	0	0	0	0	1	Clears display. Homes cursor
Return home	0	0	0	0	0	0	1	-	Returns cursor to adress 00. Resets any display shift
Set entry mode	0	0	0	0	0	1	I/D	S	I/D sets cursor move direction S sets if display shifts
Display on/off	0	0	0	0	1	D	C	B	D sets display on or off. C sets the cursor on or off B sets if the cursor blinks
Cursor/display shift	0	0	0	1	S/C	R/L	-	-	S/C sets is the screen or cursor is shifted. R/L sets direction
Function set	0	0	1	DL	N	F	-	-	DL sets 4 or 8 wire mode. N sets 1 or 2 line mode, and F sets font size.
Set CGRAM address	0	1	A	A	A	A	A	A	Sets address in CGRAM
Set DDRAM address	1	A	A	A	A	A	A	A	Sets address in DDRAM (Cursor position)

Setting	Action when set to 0	Action when set to 1
I/D	Cursor moves left after each letter	Cursor moves right after each letter
S	Display does not shift when typing	Display does shift when typing
D	Display off	Display on
C	Cursor off	Cursor on
B	Cursor blinking off	Cursor blinking on
S/C	Shift cursor	Shift screen
R/L	Shift left	Shift right
DL	4 wire mode	8 wire mode
N	One line display	Two line display
F	Character size 5 x 8 dots	Character size 5 x 10 dots

16x2 Character LCD

Display column number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	...	39	40
Row 1 address (HEX)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	10	11	...	26	27
Row 2 address (HEX)	40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F	50	51	...	66	67

→ Linia 1 – 0x00

→ Linia 2 – 0x40